

Module 3: Maximum Likelihood and Parameter Uncertainty

INFO 521 · Module 3 · Overview

Greg Chism

College of Information Science · University of Arizona

August 2026

Adapted with permission from lecture materials by Clayton Morrison, University of Arizona.

The question

Why squared error, and how sure are we about the numbers we got?

Squared error has been the loss for two modules without a reason. This module gives one: assume a noise model, write down the probability of the data, and maximise it. Least squares falls out. Then the same machinery answers a question least squares never could, which is how much your estimated weights would move if you collected the data again.

What the lectures do

Lecture A: From Loss to Likelihood

The Gaussian noise model, the likelihood and log-likelihood of a dataset, the MLE for the weights turning out to be the least-squares solution, and the noise variance getting estimated too.

Lecture B: Properties of the MLE

Uniqueness, unbiasedness and what it does not promise, the covariance of the estimator, and Fisher information as the curvature that sets it.

Appendix: The Full Covariance Derivation

The second moment, term by term.

Where this sits

This is the hinge of the course. Before it, you minimise a loss because it seems reasonable. After it, you write down a probability model and derive the loss. Every module that follows is built on likelihoods, so the derivation here is worth doing until it is automatic.

Week 3 at a glance

- Homework Unit 3, Monte Carlo and estimators.
- Checkpoint quiz 1.2 on maximum likelihood. Passing it opens Milestone 1.2.
- Milestone 1.1, least squares, is due this week.
- No activity loop this week. The parameter-uncertainty demo is used in lecture rather than as a peer loop, so the discussion carries the week on its own.
- Discussion: what likelihood buys us.
- Reading: PML Chapters 2 and 3, sections 4.1 to 4.2 and 4.7; MML Chapter 6 as review.

Watch out for

Unbiased does not mean close. An unbiased estimator is centred on the truth across imaginary repeat experiments. Your one estimate can still be far off. The covariance is what tells you how far.

The variance MLE is biased, and knowably so. It divides by N where the unbiased estimator divides by N minus the number of parameters. Lecture A shows the bias rather than hiding it.